

Fully Implicit Algorithms for Solving Partial Differential Equations

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Implicit techniques have been used in fluid flow computations for many years. Such techniques generally permit large time steps to be used in the calculation and are unconditionally stable—a distinct advantage over explicit techniques. The most commonly used implicit technique is the alternating direction implicit (ADI) method of Peaceman and Rachford [1].² ADI has been used quite successfully in both two and three-dimensional viscous flow and heat transfer problems. ADI splits the equation into a sequence of multidimensional equations, each of which is solved by the inversion of a tridiagonal matrix. Hence, in two-dimensional problems, the equation is solved in one direction (row) as a tridiagonal matrix and then solved in the other direction (column) as a tridiagonal matrix with the updated values from the first direction solution. The strongly implicit procedure (SIP) is a fully implicit technique that does not require the equation to be split as in ADI. SIP has been used by Stone [2] and Weinstein, et al. [3] in calculating transport in both two and three dimensions. Bozeman

and Dalton [4], Lin, et al. [5], Jacobs [6], and Pepper and Harris [7] have used SIP to calculate various forms of the equations of motion. Pepper and Kern [8] have also used SIP to calculate pollutant dispersion in the atmosphere.

The SIP algorithms are based on the study of Pepper and Harris [7] for two-dimensional free convective flow in a closed container. The finite difference form for vorticity, as given in [7], can be written as

$$\Omega_{i,j-1}^{n+1}B_{i,j} + \Omega_{i-1,j}^{n+1}D_{i,j} + \Omega_{i,j}^{n+1}E_{i,j} + \Omega_{i+1,j}^{n+1}F_{i,j} + \Omega_{i,j+1}^{n+1}H_{i,j} = q_{i,j}^n \quad (1)$$

where $n + 1$ denotes values at the new time steps. The coefficients $B_{i,j}$, $D_{i,j}$, $E_{i,j}$, $F_{i,j}$, $H_{i,j}$, and $q_{i,j}$ are defined as

$$B_{ij} = \frac{(-U_x)_{i,j-1}^n}{2H^2\Delta Z} - \frac{\text{Pr}}{H^2\Delta Z^2} \quad (2)$$

$$D_{ij} = \frac{(-U_x)_{i-1,j}^n}{2\Delta X} - \frac{\text{Pr}}{\Delta X^2}$$

$$E_{ij} = \frac{1}{\Delta \tau} + \frac{2\text{Pr}}{\Delta X^2} + \frac{\text{Pr}}{H^2\Delta Z^2}$$

$$H_{ij} = \frac{(U_x)_{i,j+1}^n}{2H^2\Delta Z} - \frac{\text{Pr}}{H^2\Delta Z^2}$$

$$F_{ij} = \frac{(U_x)_{i+1,j}^n}{2\Delta X} - \frac{\text{Pr}}{\Delta X^2}$$

$$q_{ij} = \frac{\Omega_{ij}^n}{\Delta \tau} + \text{Ra} \frac{\theta_{i+1,j}^n - \theta_{i-1,j}^n}{2\Delta X}$$

where a forward difference has been used for the time derivative term, and central differences have been used for the

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²Numbers in brackets designate References at end of Brief.

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spatial derivatives. The coefficient $q_{i,j}^n$ serves as an explicitly known source term to the finite difference equations. A computational molecule is shown in Fig. 1.

Equation (1) can be written in matrix notation as

$$[M] \{\phi\} = \{q\} \quad (3)$$

where $[M]$ is the coefficient matrix for the column matrix of unknown values, $\{\phi\}$, and $\{q\}$ is the column matrix containing explicitly known terms. Equation (3) is solved by altering matrix $[M]$ into a set of matrices containing only three non-zero diagonals. This creates a set of matrices that can be efficiently solved by elimination techniques.

The altered matrix is created by rewriting equation (1) as

$$\begin{aligned} & B_{i,j}\phi_{i,j-1}^{n+1} + D_{i,j}\phi_{i-1,j}^{n+1} + E_{i,j}\phi_{i,j}^{n+1} + F_{i,j}\phi_{i+1,j}^{n+1} \\ & + H_{i,j}\phi_{i,j+1}^{n+1} + C_{i,j}\{\phi_{i+1,j-1}^{n+1} - \alpha(-\phi_{i,j}^{n+1} + \phi_{i+1,j}^{n+1} \\ & + \phi_{i,j-1}^{n+1})\} + G_{i,j}\{\phi_{i-1,j+1}^{n+1} - \alpha(-\phi_{i,j}^{n+1} + \phi_{i+1,j}^{n+1} \\ & + \phi_{i-1,j}^{n+1})\} = q_{i,j}^{n+1} \quad (4) \end{aligned}$$

The first five terms on the left-hand side are the original coefficients of equation (1); the remaining two terms, $C_{i,j}$ and $G_{i,j}$, are coefficients associated with the additional unknown values $\phi_{i+1,j-1}^{n+1}$ and $\phi_{i-1,j+1}^{n+1}$, which are two new terms added to equation (1) to alter matrix $[M]$. To reduce the influence of these new terms on the altered equation, nearly equal terms (in parentheses) adjacent to the two new variables are subtracted from the new variables. The parameter, α , is allowed to vary between $0 < \alpha < 1$ to account for large variations in $\phi_{i,j}$ or widely separated grid points. Using a von Neumann error analysis in a single equation SIP model, Weinstein, et al. [3] point out that small values of α attenuate high frequency errors, and large values attenuate low frequency errors. Pepper and Harris [7] used a cyclic set of ten values for α , based on the technique used by Stone [2].

Equation (4) can likewise be written in matrix notation as

$$[M'] \{\phi\} = \{q\} \quad (5)$$

where $[M']$ now contains the seven unknown diagonal coefficients. However, $[M']$ can be factored into the product of a lower matrix $[L]$ times an upper matrix $[U]$, each containing only three nonzero diagonal elements. Fig. 3 shows the altered matrix and the two tridiagonal factored matrices $[L]$ and $[U]$. Definitions for the $[L]$ and $[U]$ diagonal terms of the algorithms used in the SIP method are

$$b_{i,j} = B_{i,j}/(1 + \alpha e_{i,j-1}) \quad (6)$$

$$c_{i,j} = D_{i,j}/(1 + \alpha f_{i-1,j}) \quad (7)$$

$$d_{i,j} = E_{i,j} + \alpha(b_{i,j}e_{i,j-1} + c_{i,j}f_{i-1,j}) - b_{i,j}f_{i,j-1} - c_{i,j}e_{i-1,j} \quad (8)$$

$$e_{i,j} = (F_{i,j} - \alpha b_{i,j}e_{i,j-1})/d_{i,j} \quad (9)$$

$$f_{i,j} = (H_{i,j} - \alpha c_{i,j}f_{i-1,j})/d_{i,j} \quad (10)$$

$C_{i,j}$ and $G_{i,j}$ are defined as

$$C_{i,j} = b_{i,j}e_{i,j-1} \quad (11)$$

$$G_{i,j} = c_{i,j}f_{i-1,j} \quad (11)$$

which are solved simultaneously with equations (6-10).

If $\{\Delta\phi^{n+1}\} = \{\phi^{n+1}\} - \{\phi^n\}$ and $[U] \{\Delta\phi^{n+1}\} = \{V\}$, equations (6-10) allow $\{V\}$ to be calculated from

$$b_{i,j}V_{i,j-1} + c_{i,j}V_{i-1,j} + d_{i,j}V_{i,j} = R_{i,j} \quad (12)$$

where

$$\begin{aligned} R_{i,j} = & q_{i,j}^n - (B_{i,j}\phi_{i,j-1}^n + D_{i,j}\phi_{i-1,j}^n + E_{i,j}\phi_{i,j}^n \\ & + F_{i,j}\phi_{i+1,j}^n + H_{i,j}\phi_{i,j+1}^n) \end{aligned} \quad (13)$$

Equations (12) and (13) are used to determine $\{\Delta\phi_{i,j}^{n+1}\}$ as

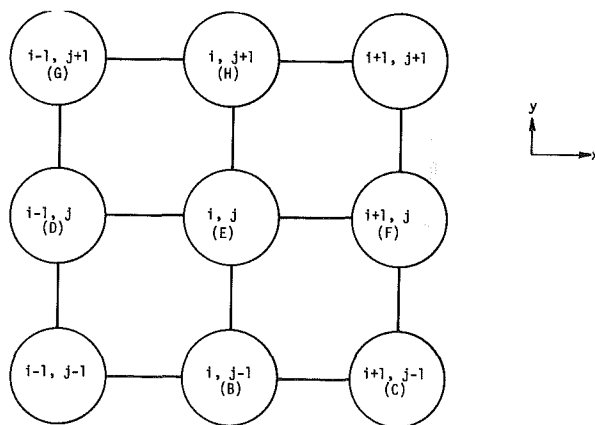


Fig. 1 Computational molecule

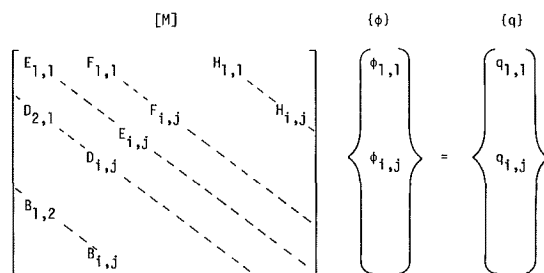


Fig. 2 Original five-diagonal matrix $[M]$

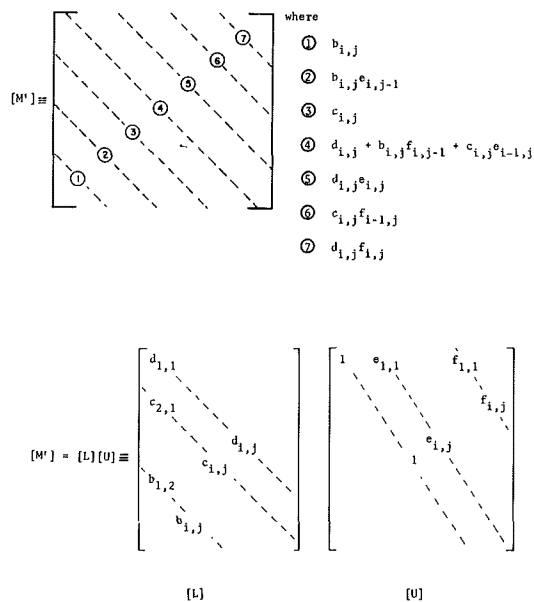


Fig. 3 Altered coefficient matrix $[M']$

$$\Delta\phi_{i,j}^{n+1} + e_{i,j}\Delta\phi_{i+1,j}^{n+1} + f_{i,j}\Delta\phi_{i,j+1}^{n+1} = V_{i,j} \quad (14)$$

Equation (14) is solved in order of increasing i and increasing j . Since $\{\Delta\phi_{i,j}^{n+1}\}$ is known, $\phi_{i,j}^{n+1}$ can then be solved from

$$\phi_{i,j}^{n+1} = \phi_{i,j}^n + \Delta\phi_{i,j}^{n+1} \quad (15)$$

When the time step is even-numbered, a different set of algorithms is used to calculate $\phi_{i,j}^{n+1}$. When $n = 1, 3, 5, \dots$,

values of $\phi_{i,j}^{n+1}$ are calculated in reverse order for i and j , with i decreasing from I to 0 and j decreasing from 0 to J .

This reordering of $\phi_{i,j}$ results in a new set of algorithms:

$$b_{i,j} = \frac{B_{i,j}}{1 + \alpha e_{i,j-1}} \quad (16)$$

$$c_{i,j} = \frac{F_{i,j}}{1 + \alpha f_{i+1,j}} \quad (17)$$

$$d_{i,j} = E_{i,j} + \alpha(b_{i,j}c_{i,j-1} + c_{i,j}f_{i+1,j}) - b_{i,j}f_{i,j-1} - c_{i,j}e_{i+1,j} \quad (18)$$

$$e_{i,j} = \frac{D_{i,j} - \alpha b_{i,j}e_{i,j-1}}{d_{i,j}} \quad (19)$$

$$f_{i,j} = \frac{H_{i,j} - \alpha c_{i,j}f_{i+1,j}}{d_{i,j}} \quad (20)$$

Equations (16–20) are solved in the order given. The results from equations (18–20) are used to determine $V_{i,j}$ according to

$$b_{i,j}V_{i,j-1} + c_{i,j}V_{i+1,j} + d_{i,j}V_{i,j} = R_{i,j} \quad (21)$$

which is solved in order of decreasing i and increasing j : Equations (19–21) are used to determine $\Delta\phi_{i,j}^{n+1}$ where

$$\Delta\phi_{i,j}^{n+1} + e_{i,j}\Delta\phi_{i-1,j}^{n+1} + f_{i,j}\Delta\phi_{i+1,j}^{n+1} = V_{i,j} \quad (22)$$

Equation (22) is solved in order of increasing i and decreasing j . $\phi_{i,j}$ at the $(n+1)$ th time step is given by equation (15).

The differences between the set of equations (6–14) and the set of equations (16–22) are subtle. Each set of algorithms are independent of each other and can be used by themselves in calculating a differential equation. However, the coupling of both sets of algorithms into an alternating sequence (but not time splitting) of even and odd time steps improves convergence rate of SIP over other implicit schemes. The accuracy of the scheme can be improved by writing equations (1–2) in Crank-Nicolson form making the scheme semi-implicit [9]. SIP also becomes a very effective scheme for solving steady-state problems when superscript n is used to denote an iteration sequence.

To compare the effectiveness of SIP with other numerical schemes, the Laplace equation for steady-state heat conduction in a rectangle was solved. Fig. 4 shows the initial and boundary conditions of the problem. The internal temperature should converge to match the boundary conditions. A comparison of the number of iterations and computing time (CPU) is shown in Table 1. The successive over-relaxation method (SOR) required the greatest number of iterations and CPU to converge to a residual of 10^{-4} ($T^{n+1} - T^n = 10^{-4}$). An optimal relaxation parameter, based on a 20×25 mesh, was used to enhance convergence. The alternating direction explicit (ADE) method required 180 iterations to converge. However, ADE required only 6 s CPU because a tridiagonal solution is not used in the solution procedure. ADI required 99 iterations and 7 s CPU. The tridiagonal matrices were solved by the Thomas algorithm. A more thorough comparison of ADI, SOR, and line iteration techniques is given by Nallasamy and Prasad [10]. SIP without the alternating sequence required 31 iterations with 5 s CPU. The alternating sequence improved the speed by approximately 20 percent.

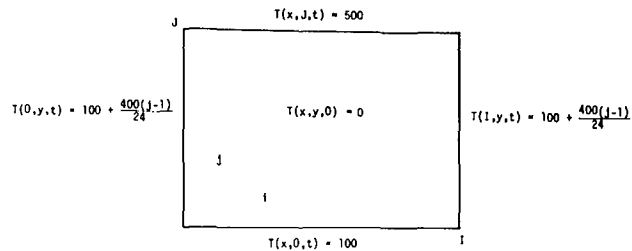


Fig. 4 Initial and boundary conditions for $\nabla^2 T = 0$

Table 1 Comparison of schemes for solving $\nabla^2 T = 0$

	SOR	ADE	ADI	SIP	SIP (ALT)
No. of iterations	540	180	99	31	25
Computation time (sec)	14	6	7	5	4

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